

Eigenvalues and inverse problems

A real-valued symbol on a Jordan curve and real Toeplitz spectra

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Topic: Spectra of finite banded Toeplitz matrices.

Rating rationale: Challenging reflects recovering a spectral implication after a gap in its original analytic proof; community impact is a structural criterion for real spectra of banded Toeplitz sections.

Note: The limiting-spectrum form of the same implication also fails; see the strengthening below.

Resolution — 2026-09-11

Negative resolution by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. [Complete manuscript](#) · [PDF](#) · [LaTeX](#). **Theorem 1 and equations (1)–(8).**

The integer-coefficient Laurent polynomial in Theorem 1 is real on a rigorously constructed star-shaped Jordan curve enclosing zero, while its 2×2 Toeplitz section has eigenvalues $-128 \pm 8i$. The proof checks continuity, injectivity and reality on the entire curve. It refutes the conjectured implication for every finite section; it does not refute the distinct limiting-spectrum statement.

The complete argument received an independent Codex-agent **PASS** on 11 September 2026. The [review report](#) records the exact scope and a hash of the original reviewed manuscript. The mathematical sections remain unchanged in the authored version. Original ChatGPT generation is disclosed. That dated informal review did not establish a formal proof certificate; the separate Lean verification is recorded below. No external human peer review is claimed. [Submission and verification record](#).

The original statement, source references and prior audit notes are retained; the former difficulty rating is historical.

Lean proof and verification evidence — 2026-09-13

Formalization: **George Stepaniants**, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA. Original mathematical proof credit remains Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge.

The [immutable complete proof](#) uses Lean 4.33.1, LeanCert 621a43d7cf21f87872392a01e874f2f1dbddc926, and Mathlib 0df444a360eaa60ab8c11dca51a86af692955474. The [project guide](#), [exact targets](#), and [source correspondence](#) identify all twenty exports and dependency pins.

`NLA.SP06.witness_counterexample` constructs the actual finite Laurent polynomial, proves a continuous injective map of the entire complex unit circle avoiding zero, proves the symbol real at every point of its image, and exhibits a nonreal point of the actual complex spectrum of its order-two Toeplitz section. `NLA.SP06.not_targetImplication` negates the complete original universal implication. The real Jordan curve is proved without assuming an existence or continuity bridge. No restricted numerical sample replaces the curve.

The repository ran [successful Ubuntu verification](#) on the pinned proof revision. All twenty Comparator statements, the permitted axiom closure and Lean’s default kernel passed, together with sandbox and rejection controls. Only `propext`, `Classical.choice` and `Quot.sound` are permitted; the [per-export axiom report](#) and [raw Linux evidence, independent audit and reproduction commands](#) are retained. Two statement reviews preceded proof implementation, and two independent final mathematical reviews passed. AI assistance and agent review are disclosed; no human peer review or Tau Ceti endorsement is claimed.

Strengthening — the limiting-spectrum implication also fails, 2026-09-18

Strengthening of the negative resolution, by Fabian Rohner and Clemens Thalhammer, Seminar for Applied Mathematics, ETH Zurich. [PDF](#) · [LaTeX](#). **Theorem 1.3 and Proposition 4.1.**

The resolution above refutes the displayed target for finite sections and notes that the distinct limiting-spectrum statement is untouched. That weaker statement also fails. The integer-coefficient Laurent polynomial $b = h^2$, $h(z) = z^2 + 97z + 103 - 10201/z$, is real-valued on a rigorously constructed real-analytic Jordan curve with winding number one about the origin, on which b takes values in $[-4000000, 0]$. Near $\lambda_0 = (20192/97)^2$ the Schmidt–Spitzer set $\Lambda(b)$ is a regular real-analytic arc crossing the real axis transversely, so $\Lambda(b) \not\subset \mathbb{R}$. Proposition 4.1 gives a four-diagonal symbol with the explicit nonreal point $(20192/97)i \in \Lambda(a)$. Since real spectra of all finite sections would force a real limiting set, this implication is weaker than the one refuted on 2026-09-11, and its failure subsumes that result.

Verification status. Author-verified preprint, with an independent Codex-agent mathematical **PASS** on 18 September 2026; the [audit report](#) records the scope, source hashes and publication corrections. No external human peer review or Lean verification of this strengthening is claimed. The counterexamples and a first draft were produced by GPT-6 Astra Pro (OpenAI) in sessions directed by F. Rohner; Opus 5 (Anthropic) produced the figures. This remains a claimed strengthening at the *Solution claimed* evidence level and does not alter the entry’s *Lean verified* status, which refers to the finite-section target.

Problem statement

Let $r, s \geq 1$ be integers and

$$b(z) = \sum_{k=-r}^s b_k z^k, \quad b_k \in \mathbb{C}, \quad b_{-r} b_s \neq 0.$$

For every $n \geq 1$, define $T_n(b) = (b_{i-j})_{i,j=1}^n$, with $b_k = 0$ outside $[-r, s]$. Suppose there is a Jordan curve $\gamma \subset \mathbb{C} \setminus \{0\}$ such that $b(z) \in \mathbb{R}$ for every $z \in \gamma$. A Jordan curve means the image of a continuous injective map from the unit circle.

Must

$$\text{spec}(T_n(b)) \subset \mathbb{R} \quad \text{for every integer } n \geq 1?$$

Why it matters

The conjecture would characterize a broad source of real spectra in nonsymmetric structured eigenvalue problems through a scalar symbol condition, uniformly over matrix size.

References

- B. Shapiro and F. Štampach, [Non-Self-Adjoint Toeplitz Matrices Whose Principal Submatrices Have Real Spectrum](#), *Constructive Approximation* 49 (2019), 191–226. Equation (2), Theorem 1(ii)–(iii), and Theorem 8; use arXiv v4, which includes the correction.
- B. Shapiro and F. Štampach, [Correction to the same article](#), 2023. The correction, pp. 27–28 of the combined arXiv PDF, withdraws the proof of (ii) $\Rightarrow$ (iii) and explicitly proposes the implication as conjectural.
- D. Giandinoto, [On reality of eigenvalues of banded block Toeplitz matrices](#), 2024, introduction and §2. The paper describes the corrected scalar implication as a conjecture before considering a block generalization.

Earlier status check — 2026-09-08

On 2026-09-08, checked the corrected arXiv v4 (2022-12-29), its appended erratum, and Giandinoto's November 2024 paper. Searches combined "Shapiro", "Štampach", "Toeplitz", "reality", "erratum", "Jordan curve", "proof", and 2026. No general resolution was located. The original article's theorem label is misleading without its correction: the finite-spectrum implication above remains the authors' stronger proposed statement. Its weaker limiting-spectrum consequence and particular symbol families are not separate catalog problems here.

Audit update — 2026-09-10

Rechecked the appended erratum in [Shapiro–Štampach, version 4](#), printed p. 27. It withdraws the relevant implication's proof and retains it as a conjecture, so the original theorem wording must not be treated as a solution. Jordan-curve/Toeplitz searches and [Giandinoto's later work](#) did not identify a proof of this scalar sufficiency implication.